



Student Result Management System

A simple and user-friendly web application built with **Python Flask** for managing student records and analyzing their academic performance. The system allows users to add, update, view, and export student results while automatically calculating totals, averages, grades, and generating reports.



Features



Student Management

- Add new student records.
- Update existing student information.
- Store data permanently using a JSON file.



Automatic Result Analysis

- Calculate total marks.
- Compute average percentage.
- Assign grades automatically based on performance.



Dashboard Statistics

- Total number of students.
- Class average.
- Top-performing student and score.



Report Generation

- Overall class report.
- Subject-wise average marks.
- Pass and fail count.
- Grade distribution.
- Individual student performance reports with remarks.



Export Results

- Export student data to an Excel file (`students_results.xlsx`).



REST API Support

The application provides APIs for:

- Viewing all students.
- Adding new students.
- Updating student records.

- Accessing dashboard statistics.
- Generating reports.
- Exporting results to Excel.

Technologies Used

- Python 3
- Flask
- Pandas
- OpenPyXL
- JSON
- HTML (Frontend Template)

Project Structure

```
Project Folder
|
├─ app.py           # Main Flask application
├─ analyzer.py      # Student and ResultManager classes
├─ data.json         # Stores student records
├─ templates/
│   └─ index.html   # Frontend page
├─ requirements.txt
└─ README.md
```

Installation

1. Clone the Repository

```
git clone <repository-url>
cd <project-folder>
```

2. Install Required Packages

```
pip install flask pandas openpyxl
```

3. Run the Application

```
python app.py
```

The server will start on:

```
http://127.0.0.1:5000
```

Open the above URL in your browser.

Grade Criteria

Average Marks	Grade
90 and above	A+
80 – 89	A
70 – 79	B
60 – 69	C
50 – 59	D
Below 50	F

API Endpoints

Get All Students

```
GET /api/students
```

Add a Student

```
POST /api/students
```

Update Student Details

```
PUT /api/students/<student_id>
```

Get Dashboard Statistics

```
GET /api/dashboard
```

Generate Overall Report

```
GET /api/report
```

Generate Individual Student Report

```
GET /api/report/<student_id>
```

Export Results to Excel

```
GET /api/export
```



Sample Individual Report

```
=====
STUDENT PERFORMANCE REPORT
=====
ID    : S101
Name  : John Doe
-----
Mathematics      : 92
Physics          : 88
Chemistry        : 90
-----
Total Score : 270
Average     : 90.0%
Final Grade  : A+
=====
Remarks: Excellent performance! Keep it up.
```



Future Enhancements

- Search students by ID or name.

- Delete student records.
 - User authentication and login.
 - Graphical analytics and charts.
 - Database integration with MySQL or MongoDB.
 - PDF report generation.
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Author

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This project was developed to simplify student result management and provide meaningful insights into academic performance through automated analysis and reporting.